E-MOSFET AS A DIGITAL SWITCH

E-MOSFET APPLICATION

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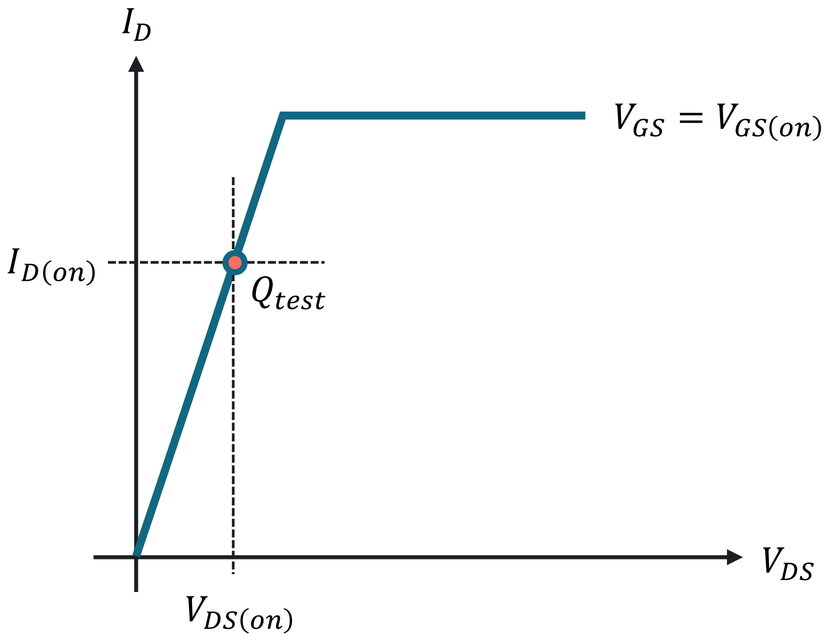
TOPIC OUTLINE

The Ohmic Region

Active-load Switching

OHMIC REGION

Measuring $R_{DS(on)}$



Drain-Source ON Resistance

$$\boldsymbol{R_{DS(on)} = \frac{V_{DS(on)}}{I_{D(on)}}}$$

For instance, at the test point, a VN2406L has $V_{DS(on)} = 1\text{V}$ and $I_{D(on)} = 100mA$.

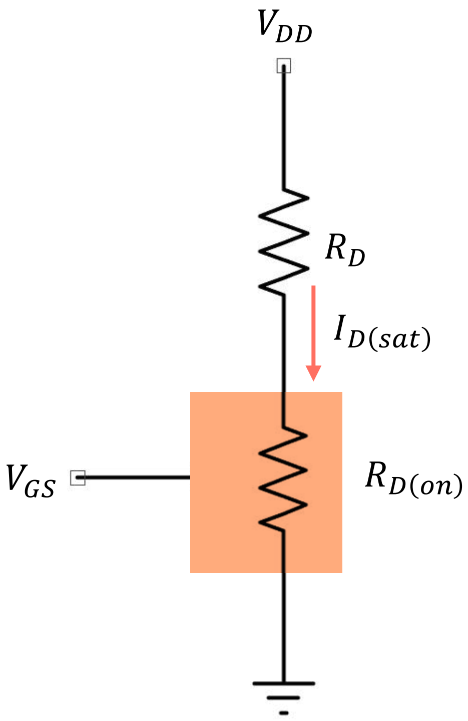
$$R_{DS(on)} = \frac{1\text{V}}{100\ mA}$$

$$R_{DS(on)} = 10\Omega$$

SMALL-SIGNAL E-MOSFETs

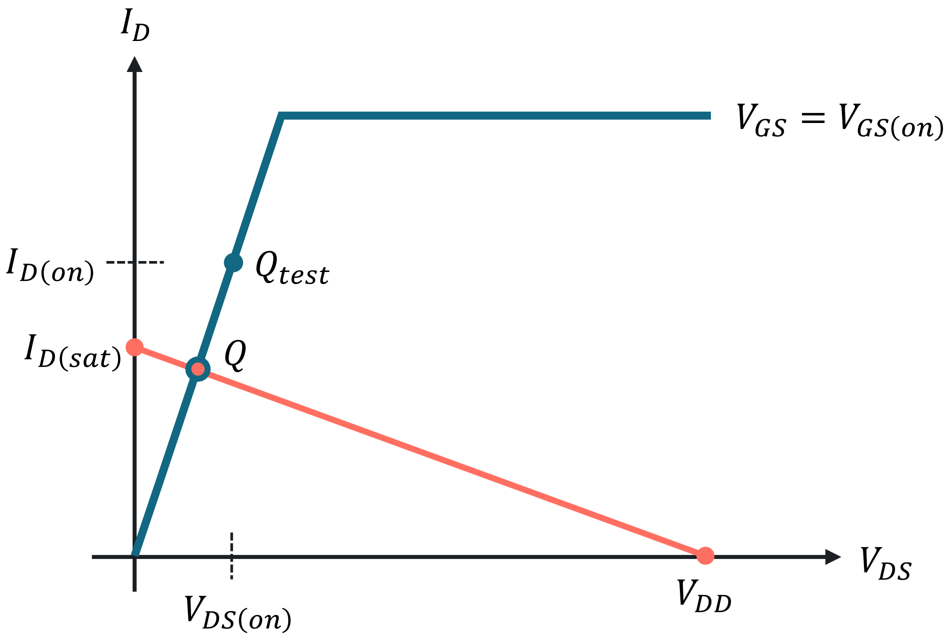
Device	$V_{GS(th)}$	$V_{GS(on)}$	$I_{D(on)}$	$R_{DS(on)}$	$I_{D(max)}$	$P_{D(max)}$
VN2406L	1.5V	2.5V	100 mA	10Ω	200mA	350mW
BS107	1.75V	2.6V	20 mA	28Ω	250mA	350mW
2N7000	2V	4.5V	75 mA	6Ω	200mA	350mW
VN10LM	2.5V	5V	200 mA	7.5Ω	300mA	1W
MPF930	2.5V	10V	1 A	0.9Ω	2A	1W
IRFD120	3V	10V	600 mA	0.3Ω	1.3A	1W

BIASING IN OHMIC REGION



Drain saturation current

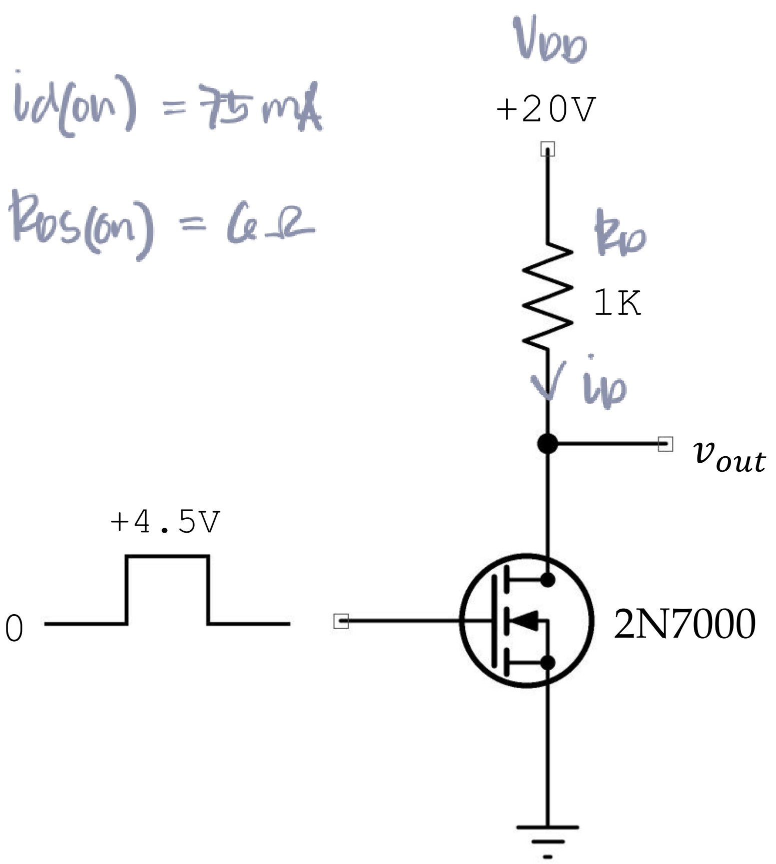
$$i_{D(sat)} = \frac{V_{DD}}{R_D}$$



$I_{D(sat)}$ less than $I_{D(on)}$ with $V_{GS} = V_{GS(on)}$ ensures saturation.

EXERCISE

What is the output voltage in the given circuit?



Solution

Check if in saturation

$$i_D(SAT) = \frac{V_{DD}}{R_D}$$

$$i_D(SAT) = \frac{20}{1K}$$

$$i_D(SAT) = 20\text{ mA} < 75\text{ mA}$$

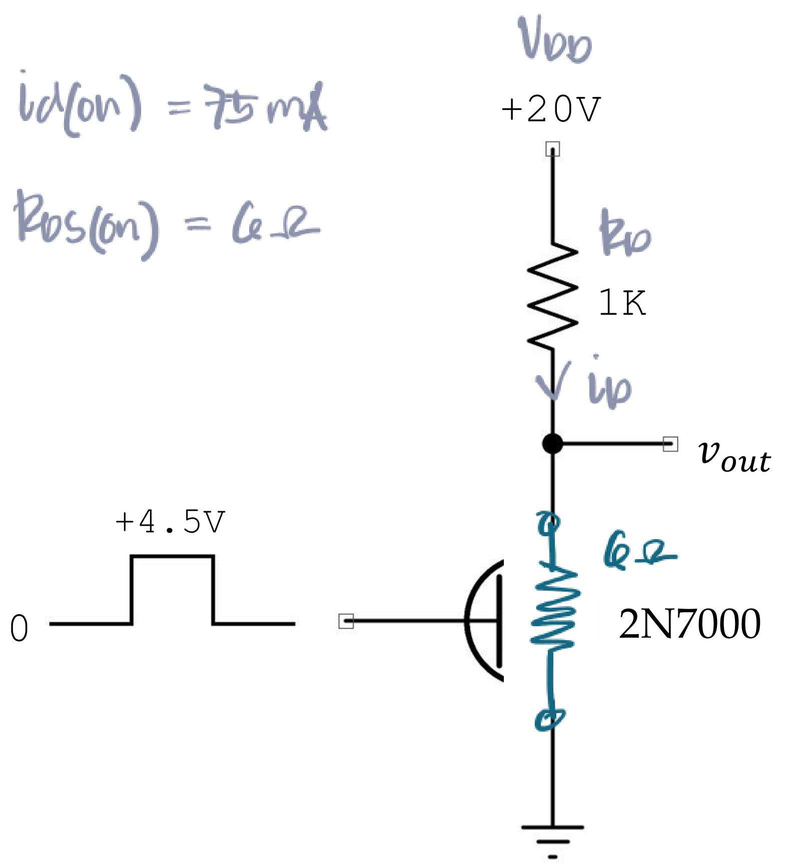
$$\therefore R_{DS(on)} = 6\,\Omega$$

EXERCISE

What is the output voltage in the given circuit?

$i_{d(on)} = 75\text{ mA}$

$R_{DS(on)} = 6\,\Omega$



Solution

Let v_{GS} = HIGH

$$V_{out} = V_{DD}\,\frac{R_{DS(on)}}{R_D + R_{DS(on)}}$$

$$V_{out} = 20\,\frac{6}{1k + 6}$$

$$\boxed{V_{out} = 0.12\text{V}}$$

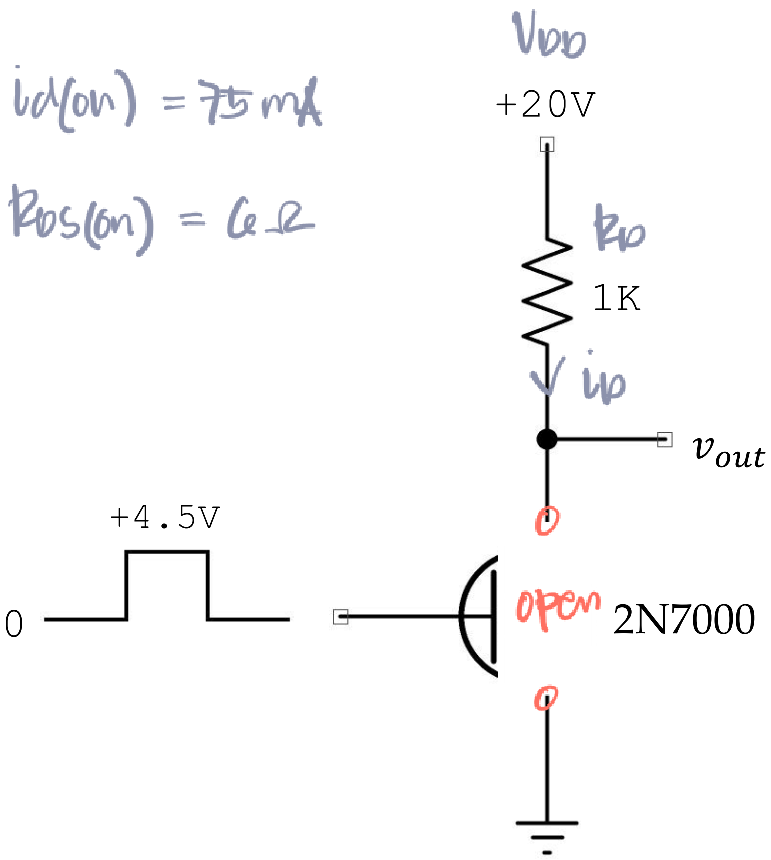
ans

EXERCISE

What is the output voltage in the given circuit?

$i_d(on) = 75\,mA$

$R_{DS}(on) = 6\,\Omega$



Solution

Let V_{GS} = Low

$$V_{out} = V_{DD}\,\frac{\cancel{R_{DS}(on)}^{\infty}}{R_D + \cancel{R_{DS}(on)}^{\infty}}$$

$$V_{out} = V_{DD}$$

$$\boxed{V_{out} = 20V}$$

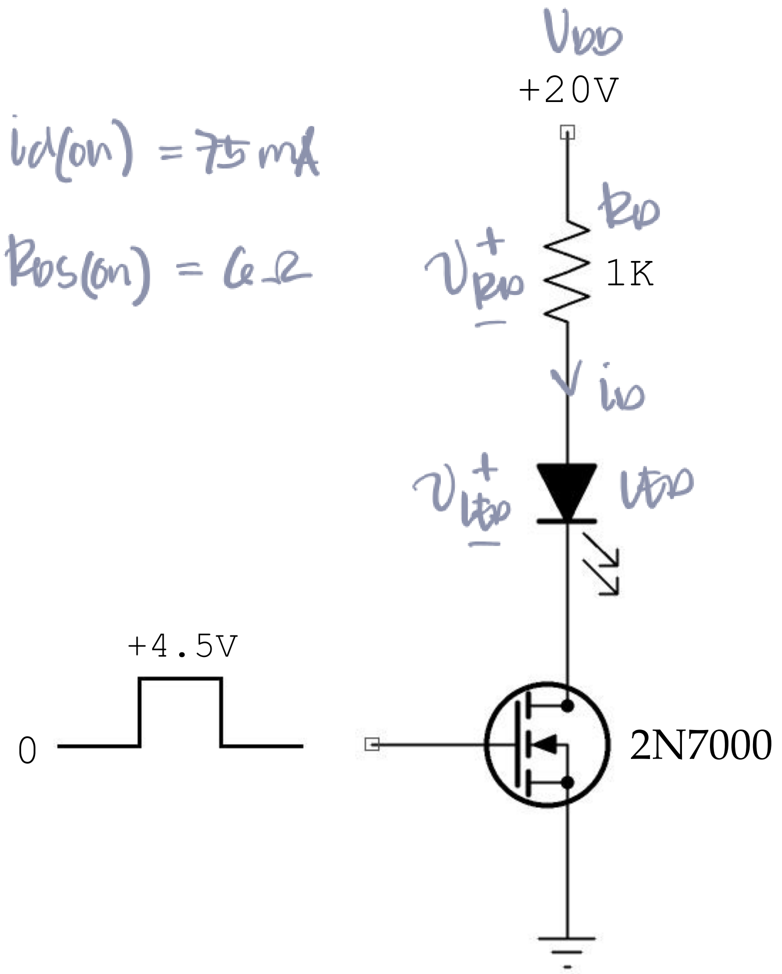
ans

EXERCISE

Calculate the LED current in the given circuit.

$i_D(on) = 75\,mA$

$R_{DS}(on) = 6\,\Omega$



Solution

Check if in saturation

$$i_D(SAT) = \frac{V_{DD}}{R_D}$$

$$i_D(SAT) = \frac{20}{1K}$$

$$i_D(SAT) = 20\,mA < 75\,mA$$

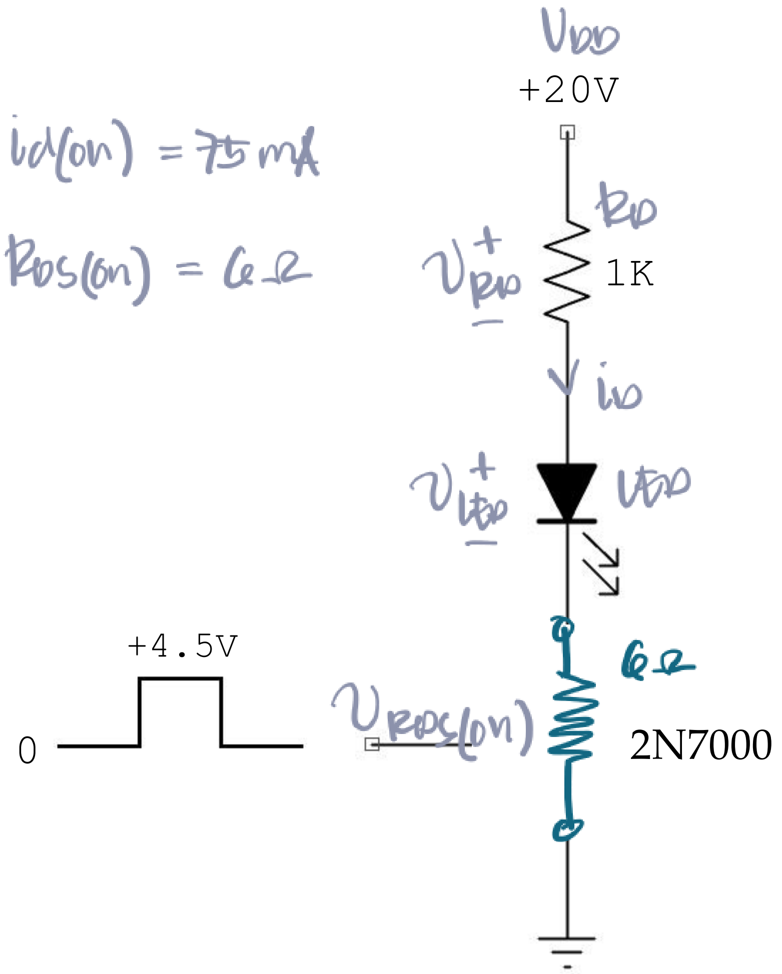
$$\therefore R_{DS}(on) = 6\,\Omega$$

EXERCISE

Calculate the LED current in the given circuit.

$i_{D(on)} = 75\text{ mA}$

$R_{DS(on)} = 6\Omega$



Solution

KVL @ D-S

$$-V_{DD} + v_{RD} + v_{LED} + v_{RD(on)} = 0$$

$$v_{RD} + v_{RDS(on)} = V_{DD} - v_{LED}$$

$$i_D R_D + i_D R_{DS(on)} = V_{DD} - v_{LED}$$

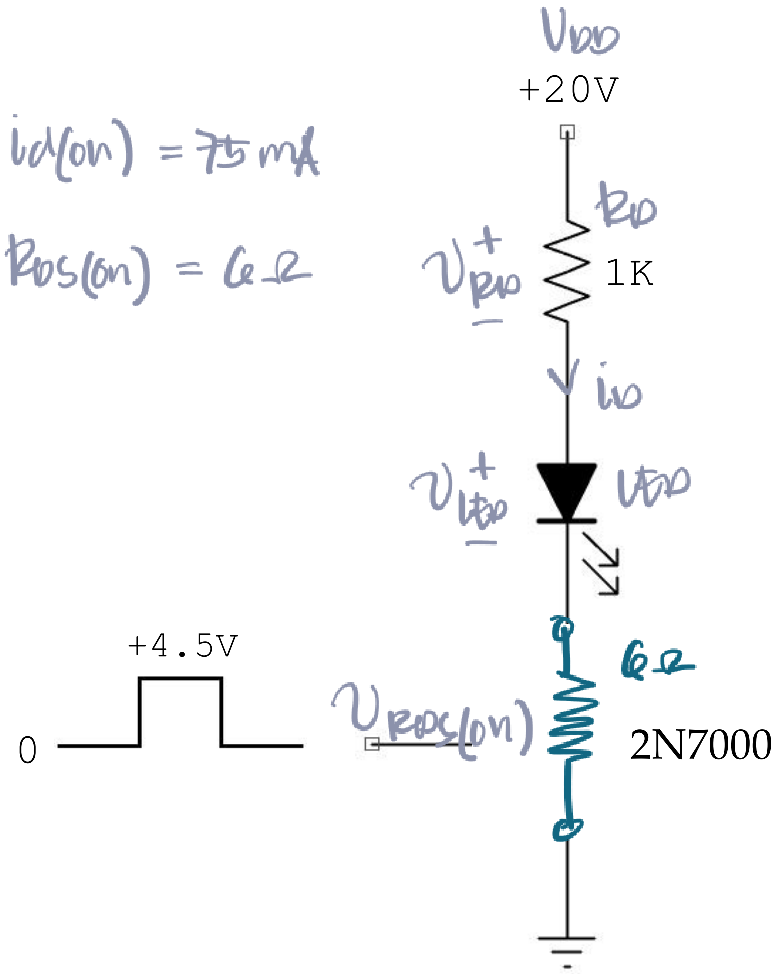
$$\frac{i_D\left(\cancel{R_D + R_{DS(on)}}\right)}{\cancel{R_D + R_{DS(on)}}} = \frac{V_{DD} - \cancel{v_{LED}}}{R_D + R_{DS(on)}}$$

1

let 2V

EXERCISE

Calculate the LED current in the given circuit.



Solution

KVL @ D-S

$$i_D = \frac{20-2}{1k + 6}$$

$$\boxed{i_D = 17.89 \text{ mA}}$$

ans

note : This is true if V_{GS} = HIGH.

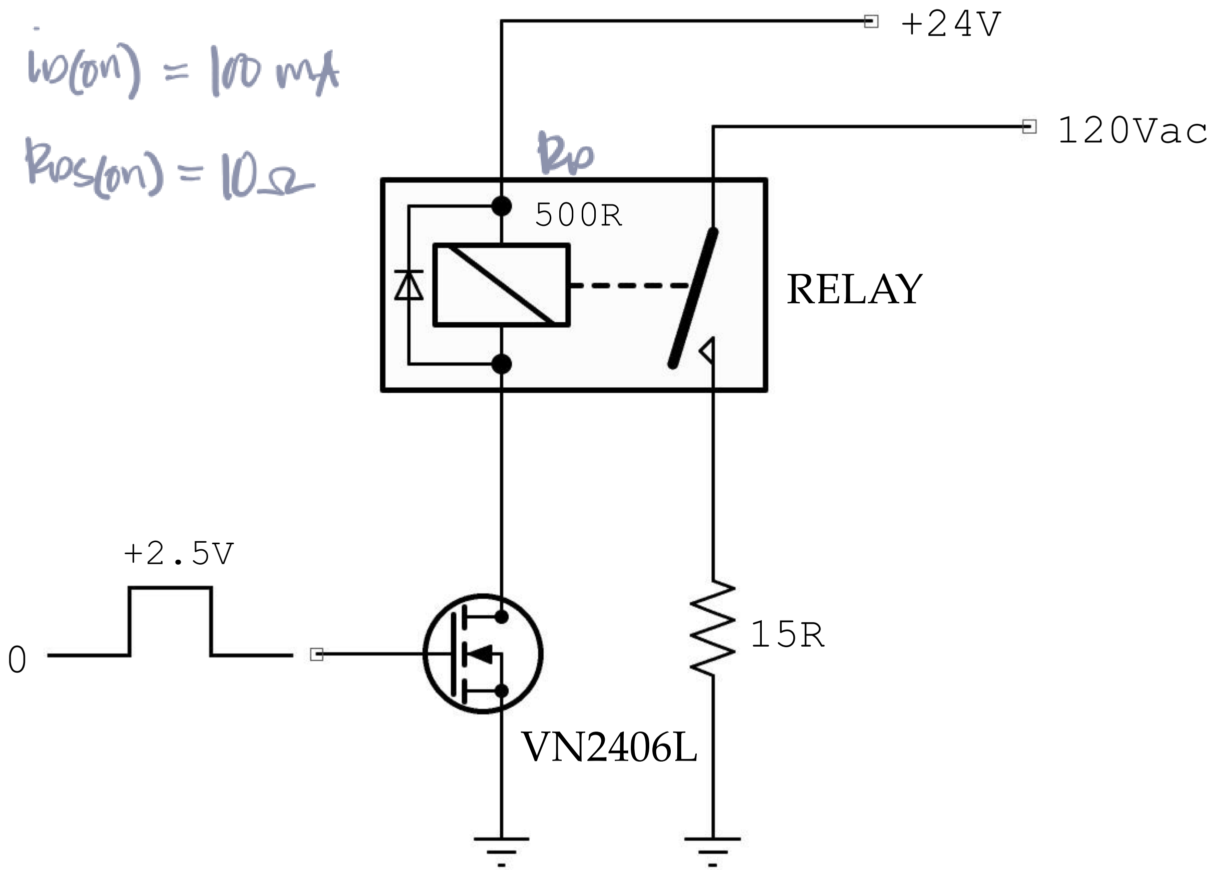
If V_{GS} = LOW, $i_d = 0$.

EXERCISE

What does the given circuit do if a coil current of 30mA or more closes the relay contact?

$i_{D(on)} = 100\ mA$

$R_{DS(on)} = 10\,\Omega$



Solution

Check if in saturation

$$i_{D(SAT)} = \frac{V_{DD}}{R_D}$$

$$i_{D(SAT)} = \frac{24}{500}$$

$$i_{D(SAT)} = 48\,mA < 100\ mA$$

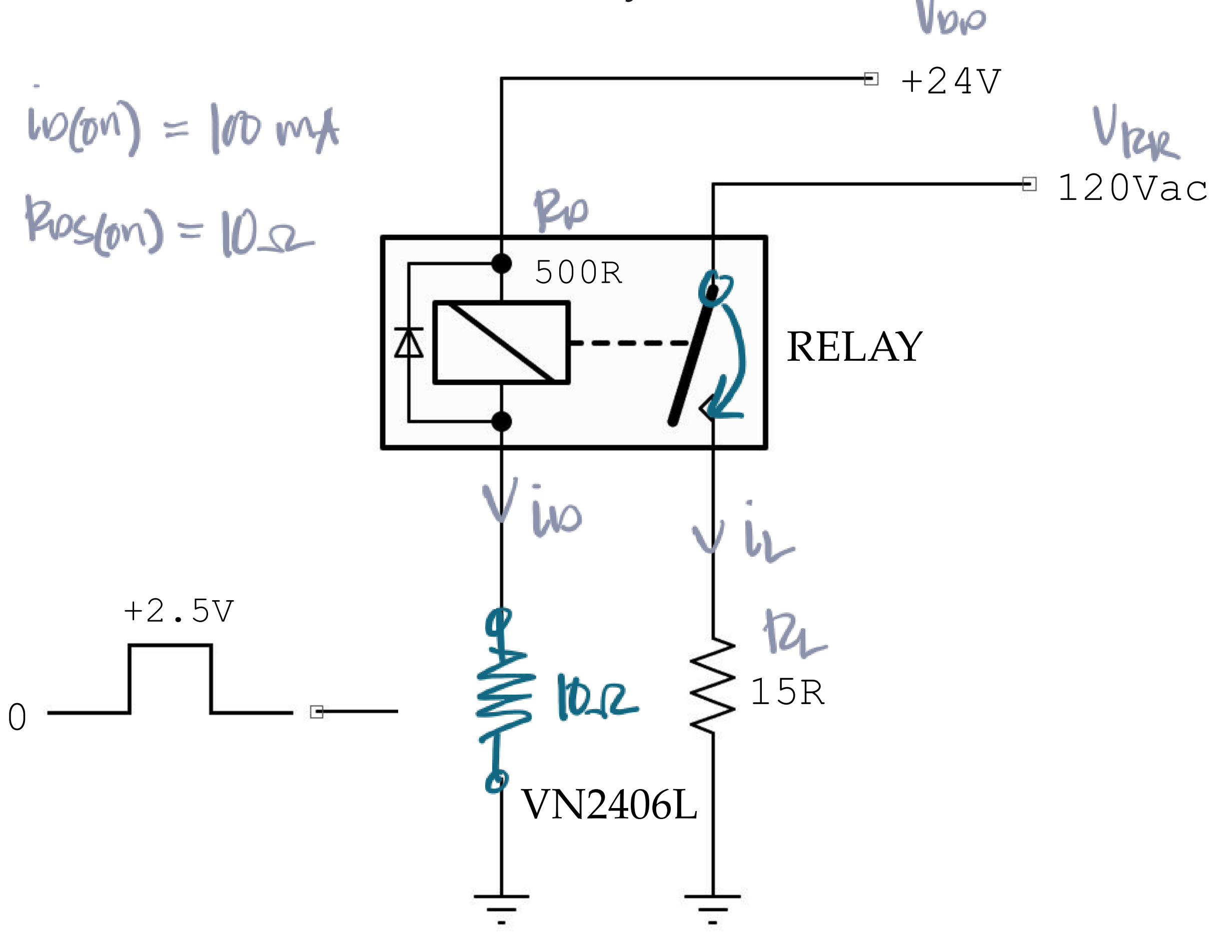
$$\therefore\ R_{DS(on)} = 10\,\Omega$$

EXERCISE

What does the given circuit do if a coil current of 30mA or more closes the relay contact?

$i_D(on) = 100\ mA$

$R_{DS}(on) = 10\Omega$



Solution

Drain Current

$$i_D = \frac{V_{DD}}{R_D + R_{DS}(on)}$$

$$i_D = \frac{24}{500 + 10}$$

$$\boxed{i_D = 47.06\ mA}$$

ans

Load Current

$$i_L = \frac{V_{RR}}{R_L}$$

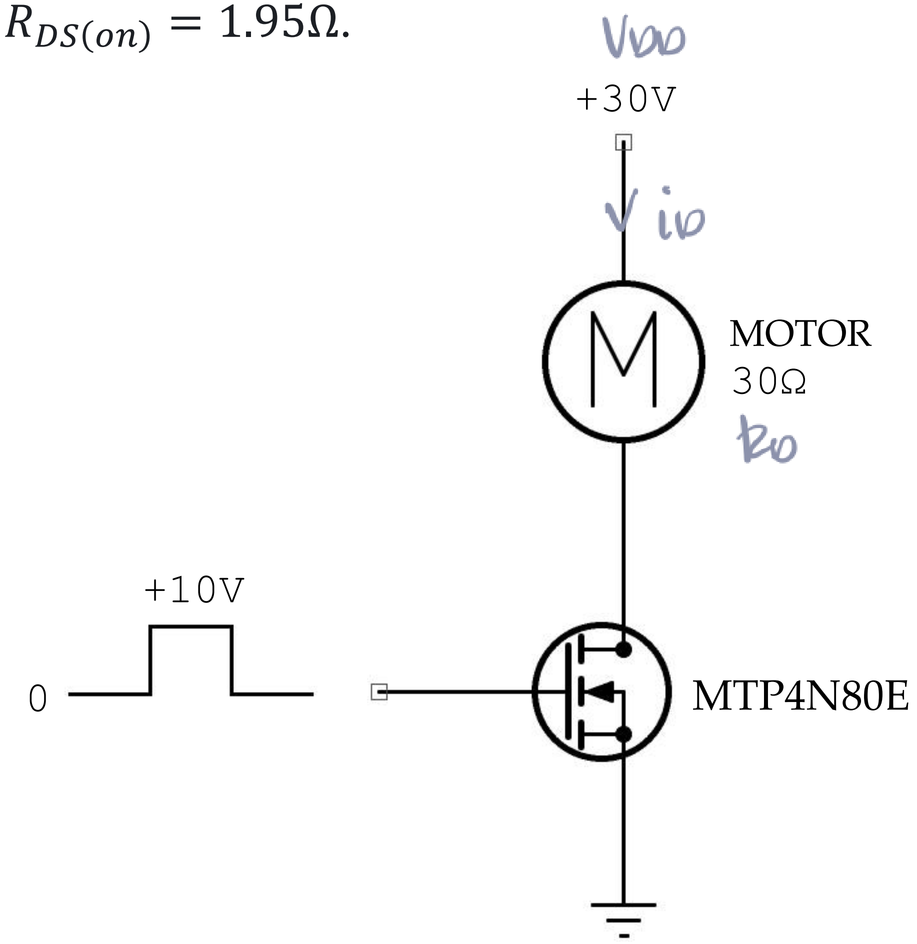
$$i_L = \frac{120}{15}$$

$$\boxed{i_L = 8\ mA}$$

ans

EXERCISE

What is the current through the motor winding? For an MTP4N80E: $V_{GS(on)} = 10\text{ V}, I_{D(on)} = 2\text{A}$, and $R_{DS(on)} = 1.95\Omega$.



Solution

Check if in saturation

$$i_{D(SAT)} = \frac{V_{DD}}{R_D}$$

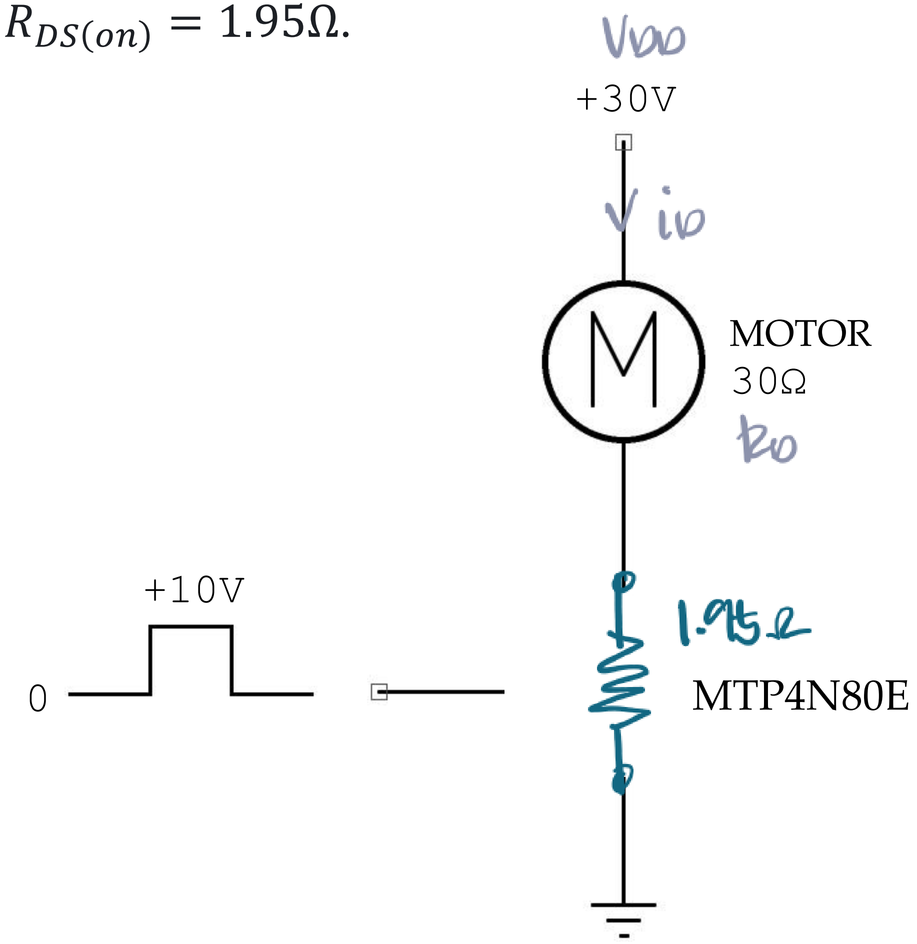
$$i_{D(SAT)} = \frac{30}{30}$$

$$i_{D(SAT)} = 1A < i_{D(on)} = 2A$$

$$\therefore R_{DS(on)} = 1.95\Omega$$

EXERCISE

What is the current through the motor winding? For an MTP4N80E: $V_{GS(on)} = 10\text{ V}, I_{D(on)} = 2\text{A}$, and $R_{DS(on)} = 1.95\Omega$.



Solution

Drain Current

$$i_D = \frac{V_{DD}}{R_D + R_{DS(on)}}$$

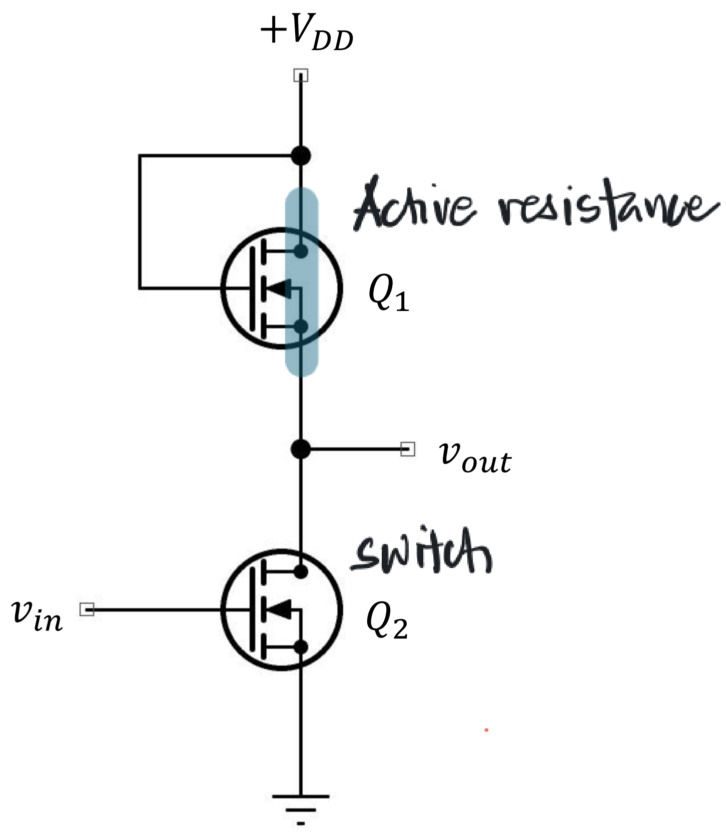
$$i_D = \frac{30}{30 + 1.95}$$

$$\boxed{i_D = 938.97\text{ mA}}$$

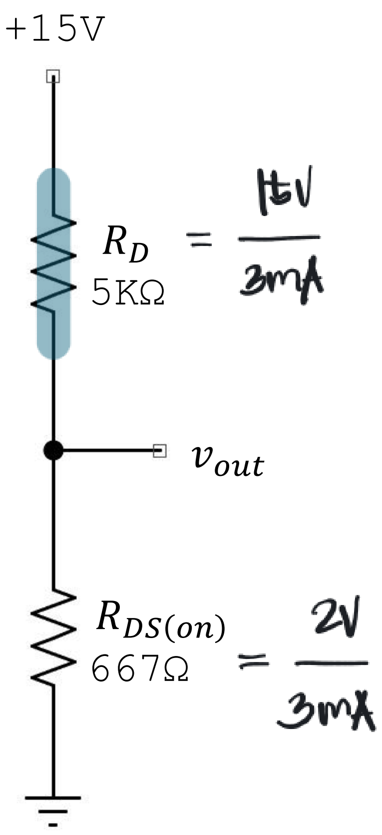
ans

ACTIVE-LOAD SWITCHING

ACTIVE-LOAD SWITCHING



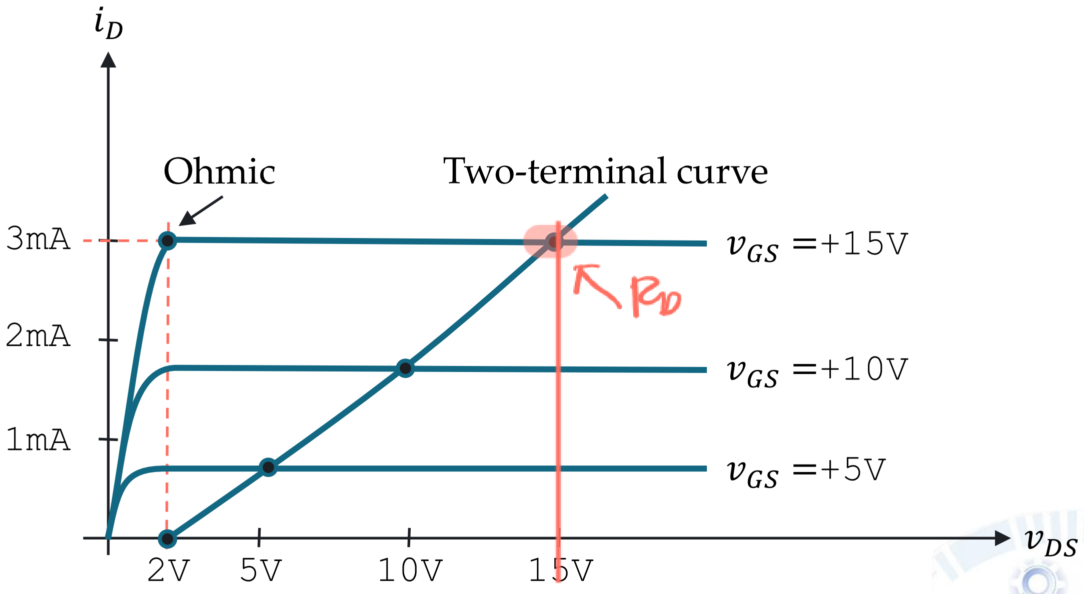
Active Load



Equivalent Circuit

$V_{GS} = V_{DS}$ produces a two-terminal curve with an active resistance of:

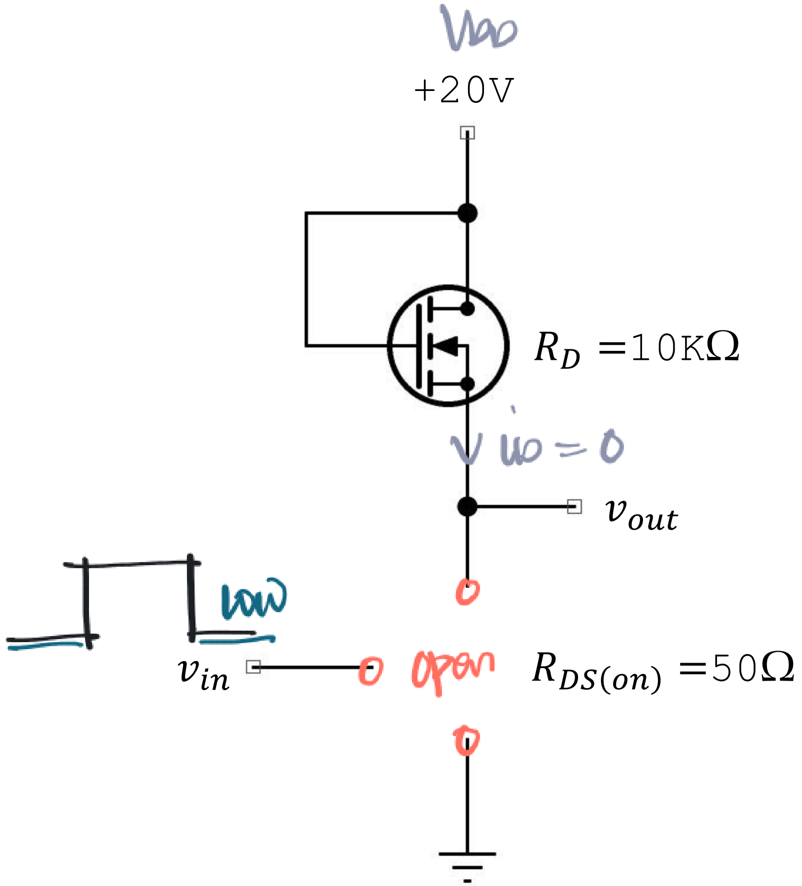
$$R_D = \frac{V_{DS(active)}}{I_{D(active)}}$$



EXERCISE

What is the output voltage when the input is low?

When the input is high?



Solution

When v_{in} = LOW

$$\boxed{V_{out} = 20V}$$

ans

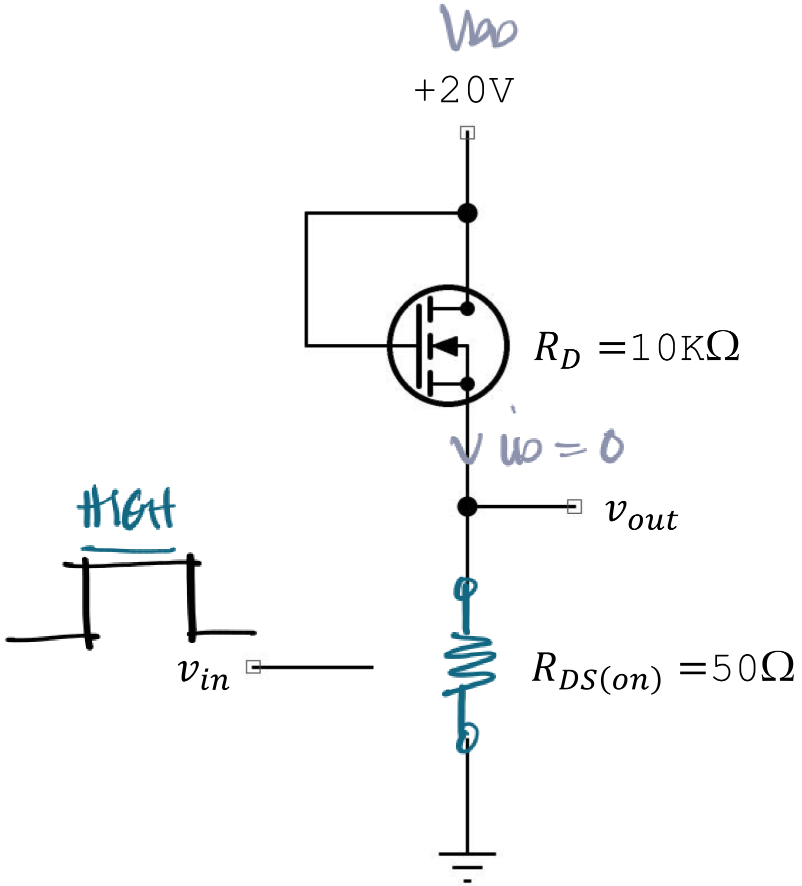
Output is pull-up to the supply voltage.

input = LOW
output = HIGH
} inverter

EXERCISE

What is the output voltage when the input is low?

When the input is high?



Solution

When Vin = HIGH

$$V_{out} = V_{DD} \frac{R_{DS(on)}}{R_D + R_{DS(on)}}$$

$$V_{out} = 20 \frac{50}{10K + 50}$$

$$\boxed{V_{out} = 99.5 \text{ mV}}$$

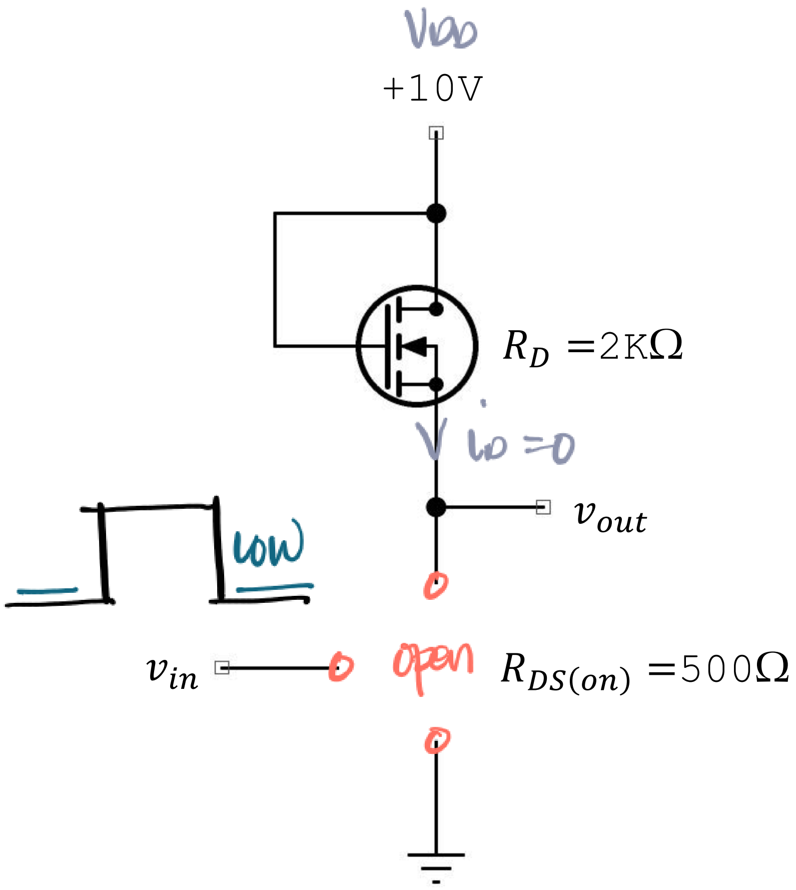
ans

Output is pull-down to ground

EXERCISE

What is the output voltage when the input is low?

When the input is high?



Solution

When v_{in} = low

$$\boxed{V_{out} = 10V}$$

ans

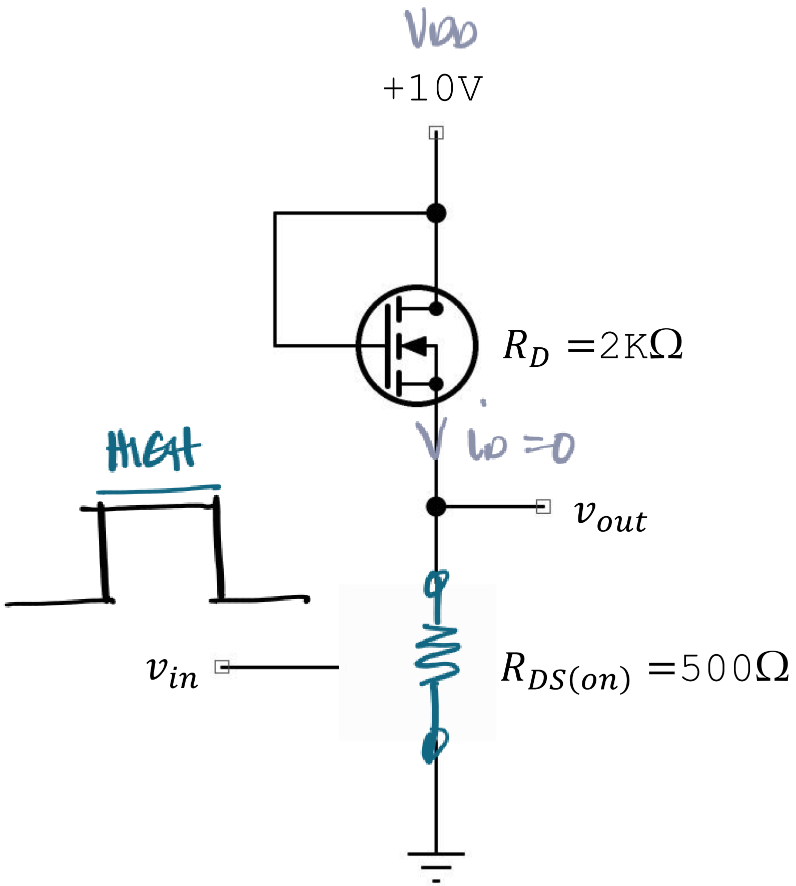
Output is pull-up to the supply voltage

input = low
output = high
} inverter

EXERCISE

What is the output voltage when the input is low?

When the input is high?



Solution

When Vin = HIGH

$$V_{OUT} = V_{DD} \frac{R_{DS(on)}}{R_D + R_{DS(on)}}$$

$$V_{out} = 10 \frac{500}{2K + 500}$$

$$\boxed{V_{out} = 2V}$$

ans

Output is pull-down to ground.

LABORATORY